# What self-driving companies should learn from drone remote operations

Mary Cummings
*College of Engineering & Computing*
*George Mason University*
Fairfax, VA USA
cummings@gmu.edu

***Abstract*—Self-driving vehicles have demonstrated impressive performance in structured and low-complexity environments but struggle in uncertain and dynamic situations. Consequently, deployed self-driving systems rely on human oversight through remote operations. This paper argues that remote operations are a safety-critical capability whose design will strongly influence the near-term viability of self-driving vehicle deployments. Drawing on more than 35 years of U.S. military experience with unmanned aerial vehicle (UAV) remote operations, this paper synthesizes five lessons directly applicable to self-driving cars: latency constraints, the importance of human-centered workstation design, the challenges of managing operator workload, the need for systematic operator training, and the necessity of robust contingency planning. Military UAV operations demonstrate that failures in any of these areas can lead to catastrophic outcomes. Recent self-driving vehicle incidents are examined through this lens, revealing critical gaps in current industry practices and offering evidence-based guidance for safer, more resilient remote operations.**

***Keywords—self-driving, remote operations, UAV, drone, lessons***

## I. Introduction

While developers of self-driving cars have made many impressive advances, onboard artificial intelligence (AI) continues to struggle when making decisions in novel situations [2]. While such systems may perform well in low complexity scenarios and do not get distracted or fatigued like humans, scenarios with significant uncertainty often result in self-driving vehicles behaving in unpredictable ways. Recently Waymo's fleet of self-driving cars froze in place when San Francisco experienced a power outage [3]. Cruise's self-driving car operations ended when one of its vehicles could not cope with the uncertainty of a jaywalking pedestrian, causing severe injury [4]. Because of such events and gaps in self-driving vehicle capabilities, all these cars currently require some degree of human oversight and intervention in the form of remote operators who supervise and direct the cars via a digital connection.

Remote operations for unmanned vehicles have a long history. With the rise of Unmanned Aerial Vehicles (UAVs), aka drones, it was recognized by the Department of Defense (DoD) in the 1980s that humans would need to supervise drone operations through video and telemetry data links [5]. The DoD experienced a sharp learning curve for UAV remote supervision due to many accidents caused by inadequate designs of both the remote operations stations but also in the design of training programs for UAV operators. The military's more than 35 years of remote control and supervision lessons learned for UAV operations, often in blood, could inform remote operations of self-driving cars so safety can be expedited.

## II. Human Supervisory Control

Remote human oversight of autonomous vehicles has a formal name, which is human supervisory control [6]. For self-driving cars, supervisory control can happen in one of four ways, as outlined in Table 1. Teleoperation is a classic and low-level form of supervisory control where remote humans have a set of controls that allow them to directly control speed, heading and turn rates. As seen Table 1, teleoperation can be done remotely using a proxy driving simulator, but it can also be executed in or near a vehicle. As will be discussed in a later section, teleoperation is especially sensitive to any time delays of the control signals.

TABLE 1: DIFFERENT TYPES OF SUPERVISORY CONTROL IN SELF-DRIVING OPERATIONS

| Remote Driving | Remote Assistance | In-vehicle Teleoperation | In-situ Teleoperation |
|---|---|---|---|
| An operator directly makes longitudinal (acceleration or deceleration) and turning/heading changes through a dedicated communication channel. Most companies use controls seen in car racing video game simulations that include a steering wheel and brake and acceleration pedals. | An operator issues goal-based commands to a vehicle through a point-and-click paradigm to control speed and path planning, often called laying breadcrumbs. This mode also includes humans interpreting the world for the car due to computer vision limitations, e.g., hand signals and construction zone signage. | A set of physical or digital controls located *inside* the vehicle, which allows an operator to directly control speed and heading. This mode of control typically is used intermittently when the onboard autonomy cannot safely operate, such as an operator sitting in the right passenger seat. | A set of physical or digital controls located *outside* the vehicle and within line of sight, allowing for either rate control of speed & heading changes (the use of a trailing chase car to provide emergency oversight), or 2) goal-based control, similar to remote assistance, e.g., Tesla Smart Summon. |

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A more robust form of remote operations is remote assistance. By providing, for example, a breadcrumb trail of dots to follow on a screen, a remote operator can instruct a vehicle which path to follow and also specify a speed, so that the car does the actuation. While not as time sensitive as teleoperation, this mode is still affected by time delays. In-vehicle teleoperation (Table 1) typically occurs in early-stage testing for self-driving cars and in-situ teleoperation only works at slow speeds in very limited environment. Fleets are controlled through remote driving and assistance, so these modes are more aligned with extensive military UAS operations and are the focus of this paper.

## III. The Five Lessons

Since the US military has been operating with remote operations centers where humans have performed both remote teleoperation and remote assistance for 35 years, there is a large body of literature on these remote operations. In this research about military UAV remote operations, five themes are dominant: Latency, workstation design workload, training and contingency operations. The following sections detail these lessons learned.

### *A. Latency in the Elephant in the Room*

When any type of vehicle is remotely operated, latency of the command and video signals to and from the remote center is a primary concern. In the early days of UAV operations, the Air Force allowed their pilots, located in Las Vegas, to remotely take off and land using teleoperations in the middle east. With a typical delay of 2s [7], their accident rate was so high, that these operations were replaced with local operations (short range, line-of-sight) [8] and eventually completely automated [9]. Even beyond take-off and landing, the fragility of the data-link made it very difficult for aircrew to adequately perform their missions [10].

The problems with control and signal latency are exacerbated by a well-known human-induced latency called the neuromuscular lag, which means that humans cannot reliably respond to incoming information less than 200-500ms [11]. Thus, any remote operator cannot be expected to correctly respond to external stimuli that requires a response in less than that time, which have led to many UAV crashes. For self-driving teleoperation, any operations requiring a remote operator to take control in less than .2 s to avoid an accident (assuming no delay in the signal, which is never true for remote operations) is very unsafe.

In terms of signal latency, the typical approach for self-driving car companies is to use the best of three cell phone signals to transmit messages, with the best meaning the one with the shortest delay. Cell phone signals are notoriously unreliable, particularly in urban canyons and dead zones so it is generally recognized that self-driving teleoperations are not robust to signal problems. Instead, companies like Waymo prefer the remote assistance option where operators interpret signs AI cannot understand and to assist in situations of high uncertainty.

While remote assistance is more robust to latency than teleoperations, self-driving vehicles can still suffer from latency issues. A Waymo vehicle was struck broadside when a remote operator gave a command to a vehicle to make a left turn on a yellow light, which was actually red due to the delay in communications [12]. These delays are dramatically worsened by Waymo's choice to move its remote operations center from the US to the Philippines [13], which was very much on display in the San Francisco power outage in 2025. In addition to problems with remote assistance signal delays and operator neuromuscular lag, there is an inherent cognitive switching cost when remote assistance operators switch between tasks, which can cause even further delays. Such switching costs are well established in UAV operations and discussed further in the workload section.

### *B. Workstation Design Matters*

Another major lesson learned from UAV operations is that poor workstation design, both controls and interfaces, can undermine efficiency and safety. The US military has lost many UAVs because basic human factors considerations were not applied to the design of cockpit controls, decision interfaces, and issues with mode confusion. Mode confusion occurs when an operator does not understand the current mode of automation. This lack of understanding can cause catastrophic human-system failure due to confusion over who is in control (the human or system). This is a pervasive problem in manned and unmanned system, leading to UAV crashes [14, 15]. The lack of human factors considerations and finger-in-the dyke approaches to the development of procedures to help operators combat uncertainty are significantly contributors to unmanned system crashes [10, 16-19].

Fig. 1 illustrates that for the first 18 years of military UAV operations for four different platforms, human factors issues were a significant cause of crashes [1]. The Air Force's Predator had the most human factors-caused crashes (67%). These crash causes include display design, procedure design, operator fatigue and training issues. To further elucidate the problems caused by poor workstation design, Fig. 1 also depicts how many of the human factors-caused crashes were due to poor interface design (both visual and aural), as well as crashes caused by poor procedure design.

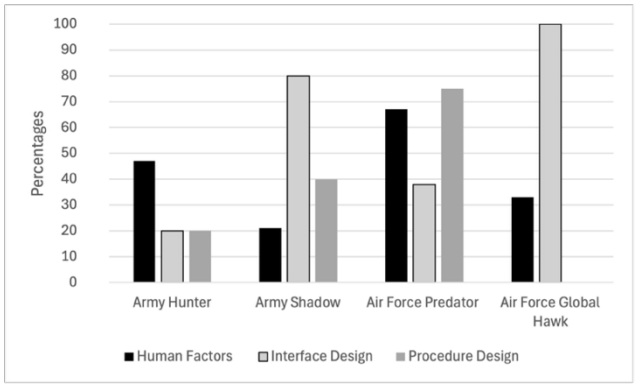


**Fig. 1. UAV crashes (1986-2004) caused by human factors issues, including poor interface and procedure design. These categories do not sum to 100% because both factors could be present in an accident [1].**

The Air Force's Global Hawk, a highly automated aircraft, only lost a single aircraft because of human factors problems, which was caused by a poor interface that allowed the aircraft to taxi at 155 kts. The Army's Shadow had a high number of crashes due to poor interface design (80%). Lastly, poor procedure design was a factor for 75% of Predator accidents, but the US Army also experienced these problems as well.

Because companies have been very opaque about self-driving remote operations, it is difficult to know whether and to what degree their designs suffer from similar problems. However, recent reports of self-driving shuttle accidents provide some insight into possible poor control designs. Many slow speed shuttles have onboard operators that use gaming devices like an Xbox or PS5 controllers to control the vehicle *in situ* (Table 1) through intersections or other complex scenarios. These are considered nonstandard devices and there is a strong likelihood there could be a negative transfer of training if gamers become shuttle operators. In addition, such devices require very little pressure to move and operators who are not secure and stable in their control positions can experience inadvertent control actuation. In the military, force feedback was inserted into digital controls forty years ago to prevent accidental misapplications. In terms of remote operations, Waymo's inability to operate during the 2025 power outage, not even to recall their vehicles, suggests there are problems in the design of their workstations.

### *C. Operator Workload*

Military UAV remote operators are often overworked and paradoxically, also suffer from intense boredom, depending on the operations. For example, if remote operators teleoperate during missions, their workload is extremely high and fatigue becomes a significant factor in crashes [20]. On the flip side, during periods of remote assistance where the onboard autonomy performs well, such as holding to wait for a person of interest to emerge from a building, boredom has been a major contributor to degraded performance [10, 21]. Vigilance, the ability to sustain attention in low workload scenarios, has been shown to cause high stress and exhaustion to remote operators of such systems. Complacency, a common outcome when vigilance is an issue, plays a large role in many operators' inability to sustain attention and has been cited in past [22] and recent UAV crashes [23].

Little is known about the workload levels of remote operators of self-driving cars, as companies have been very opaque. Their tasks appear to range from interpreting signs to assisting self-driving cars when they get stuck in a dead end and even figure out what to do when they drive into a flood zone. When the scenarios are simply and relatively straightforward, workload and stress are low. Indeed, in these scenarios, boredom and complacency may become factors. However, when scenarios are complex and real emergencies, like during a power outage or when a car drives into an active shooter scene, just like the UAV operators, self-driving remote operators can find themselves quickly overwhelmed. Research has shown that emergency remote operations for can cause remote operator workload to spike and that staffing levels under nominal conditions will likely not be sustainable under emergency conditions [24]. Waymo's inability to cope with the power outage in San Francisco further validates this concern.

Workload levels are directly tied to staffing, and companies are incentivized to maximize remote operator utility to save money. Given reports that Waymo has 70 operators on duty in the Philippines to supervise around 2000 cars [25], these remote operators are supervising around 29 cars each. Research has shown it is possible to have one person supervise multiple autonomous vehicles [26], but controlling high numbers of vehicles results in high stress and cognitive switching costs that undermine system performance [27]. This means that operators must spend additional time gaining situation awareness about the new vehicle after a switch. Chinese regulators have recognized this and have mandated that a single remote operator can only supervise three vehicles [28].

There is very little data about the performance of Waymo's remote operators but one recent stream of data provides some insight into how well remote operators are handling such high workload. Recently the San Francisco Municipal Transportation agency provided data that shows how often Waymo vehicles block city busses and trains [29]. Based on this data, Fig. 2 illustrates that since 2023, there has been a dramatic rise primarily in the number of Waymo blocking and freezing events, especially between 2024 and 2025. The near-miss and illegal behaviors for Waymo exceeded the number of collisions in 2025, which is important because such events are leading indicators of collisions.

The blocking and freezing events occur when Waymo's software is incapable of handling the current situation and these situations exemplify where a remote operator must be called to provide assistance to determine the correct course of action. Waymo expanded operations in San Francisco through 2023-2025 so the rise in events is not unexpected. To determine how well the Waymo remote operators are coping with this increased number of events, the delay times associated with these events are instructive.

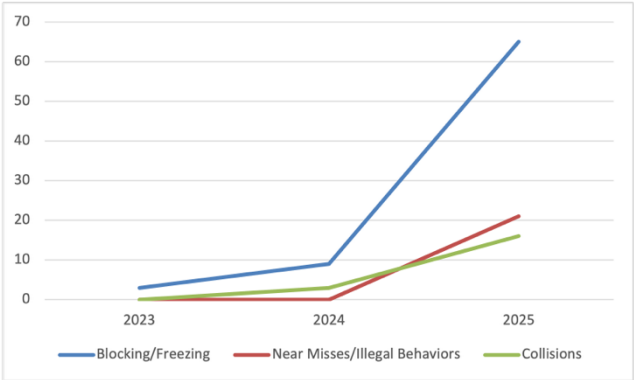


**Fig. 2. From 2023-2025, the number of events where Waymo self-driving cars froze in place or either blocked San Francisco busses and trains, engaged in near misses or other illegal behaviors, and collided with busses or trains.**

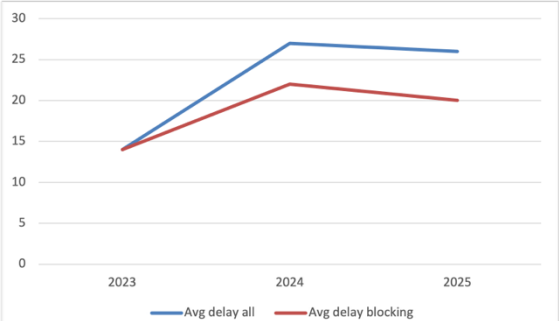


**Fig. 3. From 2023-2025, the average delay time in minutes when Waymo self-driving cars collided and froze in place or blocked San Francisco busses or trains, compared to just delay times due to freezing and blocking (no collisions).**

Figure 3 demonstrates how average delay times have changed between 2023-2025 for all incidents involving collisions and freezing and blocking events, as well as the average delay times for just blocking and freezing events [29]. As expected, when the data includes collisions, the time to clear an event for remote operators is, on-average, 5-6 minutes higher than just for blocking and freezing events. However, what is critical to note is that the blocking and freezing averages between 2024 and 2025 (M = 21.5 m, SD = 9.2 m and M = 20.3 m, SD = 15.1 m, respectively) are statistically the same ($t_{55} = -0.217$, $p = 0.82$). This means that for this one sample, delay times are not improving,

While it is a positive result that delay times are not getting worse as operations ramp up (for both numbers of vehicles and remote operators), the flat delay times suggest that there is little-to-no maturation occurring for the quality of remote operations in trying to clear delays. There should be a noticeable decline in delays over time - no congested city like San Francisco can tolerate self-driving cars blocking roads for 20 minutes at a time. The data in Figs. 2 and 3 suggest Waymo needs to take a hard look at the quality of its remote operations, especially as it relates to possible high levels of workload. The US military has been trying to invert staffing from multiple people controlling a single medium or large UAV to a single person controlling multiple UAVs since the early 2000s. Despite hundreds of millions of dollars in research and development, the U.S. military still requires multiple people to supervise a single UAV, primarily because of workload and cognitive switching costs.

### *D. Operator Training*

In a perfect world, an organization would conduct a knowledge, skills, and abilities (KSAs) assessment prior to operations in order to determine the most effective training program. Especially for vehicles controlled by autonomy and/or artificial intelligence, it is important to ensure operators are trained so that they understand the modes of autonomy and AI so that mode confusion does not result. Unfortunately, many organizations consider training as a low priority item and are more reactive instead of proactive. In terms of military UAV operations, the first formal KSA analysis was not performed until well after UAV operations were established [30]. The result of this lack of training preparation for a new field of supervisors of autonomy and AI was a string of UAV accidents [19].

Since there are no regulations regarding the training or certification of self-driving cars and companies are very secretive about their practices, little is known about the training programs for remote operators. There are no published studies about the required KSAs for remote operators of self-driving cars and there are few indications that the self-driving industry has been proactive in determining minimum training requirements. One exception is recent research by Vay remote driving technologies, who demonstrated that experience and training are critical for effective remote driving and as expected, latency can cause controllability issues [31]. Beyond this sliver of information, there is still a significant amount of needed research in determining the core KSAs for both remote assistance and driving.

Certification is also a key element of training in that operators must be certified by independent authorities to ensure the highest preparedness for operations in safety-critical environments. Military personnel are held to a strict checkride assessment policy that should be emulated in self-driving remote operations. Moreover, such checkrides also should reflect major upgrades in any system capability. For example, if a new end-to-end learning system is put in a self-driving car stack, new remote operator training programs and certifications are also needed to ensure the human supervisors are not caught off guard with new software upgrades.

### *E. Contingency Operations*

Contingency planning is a core element of both military and commercial aviation. Prior to every flight for both manned and unmanned flights, there is a brief about what to do if radios fail. Global Hawk UAVs land themselves if they lose link and other less capable UAVs fly themselves to safe areas over unpopulated terrain to attempt to reestablish contact [32]. However, the U.S. RQ-170 Sentinel downed by Iran in 2011 using GPS spoofing highlights what can happen when companies cut corners. There were no onboard protections for GPS hacking, even though this was a well-known capability at that time [33]. Cybersecurity is a critical threat to both unmanned aviation assets and self-driving cars, especially GPS spoofing, but there is no indication from self-driving companies that they are prepared for such attacks, which are not difficult to launch.

Waymo's inability to operate during the power outage in San Francisco in 2025 also signals a lack of proper contingency planning. The military addresses such possibilities by having onsite generators, as well as multiple physically disparate remote ground control stations so that if one station goes down, UAV supervision can be very quickly transferred to another remote operations center. Supposedly self-driving cars can execute a minimum risk maneuver such as pulling to the side of the road if there is a problem that prevents the car from continuing a trip. However, the power outage resulted in cars frozen in place on the road, blocking both regular traffic and first responders, so it is clear more needs to be done to ensure these minimum risk maneuvers are not performative.

## IV. Relating these Factors to Accidents

To better understand how these five factors relate to accidents for UAVs, they are placed into the Swiss Cheese accident framework (Fig. 4). The Swiss Cheese framework, originating from aircraft accident research [34], illustrates that accidents occur in complex systems when multiple layers of defense—organizational policies, supervision, working conditions, and operator actions—have weaknesses, i.e., holes, that align. These layers are notionally intended to prevent hazards from leading to harm, but failures can and do occur. These weaknesses are often systemic rather than the fault of one person, so that even when a pilot or remote operator makes a mistake, i.e., an unsafe act, failures in the other layers can significantly contribute to severe consequences from a human mistake.

Figure 4 illustrates which layers the five key considerations for effective remote operations influence in the Swiss Cheese framework. Inadequate training, poor workstation design and significant latency all are preconditions for unsafe acts for both UAVs and self-driving, as previously discussed. What is very concerning is that for self-driving remote operations, if these hazards are not explicitly addressed, it is just a matter of time before all three factors contribute to a crash.

Operator workload sits between the unsafe supervision and the preconditions for unsafe acts layer in Fig. 4. This is because in remote operator centers, there should be two layers of supervision for the humans controlling the vehicles, managers on duty whose directly oversee operations and managers who do the medium-to-long range staffing planning. The US military has spent significant money on developing both the real-time management and overarching staffing models. However, it is not clear to what degree self-driving car companies are engaging in these best practices. For example, did Waymo organically develop the practice of requiring humans to supervise 29 vehicles or was there a first-principles approach that determined this was the best approach?

At the organizational end of the Swiss Cheese framework, contingency planning sits between the unsafe supervision and organizational layers because this function should be dictated by leadership and executed by supervisors. For example, it was clear that real-time human supervision was inadequate in the massive power outage that caused havoc for self—driving cars in San Francisco, but it was also a leadership failure for Waymo.

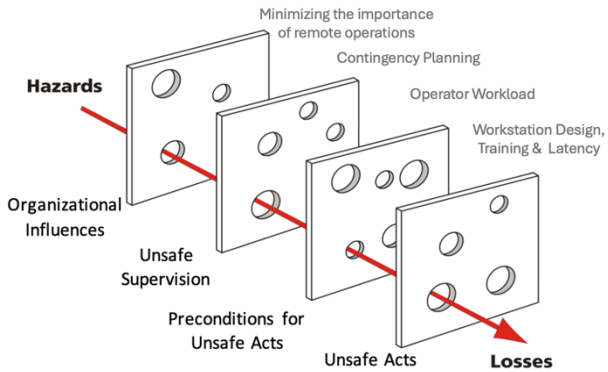


**Fig. 4. Remote Operations and the Swiss Cheese Model**

Lastly, the summative lesson that should be learned from these factors is that the importance of grounded and well-resourced remote operations cannot be trivialized. Remote operations have been the critical lynchpin for US drone successes, but many aircraft were lost until the lessons laid out in this article were learned. Self-driving car leaders should heed these lessons and embrace the fact that remote operators are not a temporary fix for artificial intelligence software problems. They are and will continue to be for some time an integral part of the self-driving ecosystem

## V. Conclusion

The history of military UAV remote operations offers a cautionary but instructive roadmap for the self-driving car industry. These lessons are unequivocal: minimizing latency is paramount, poor workstation design amplifies human error, operator workload can fluctuate between dangerous overload and performance-degrading boredom, inadequate training can lead to crashes, and insufficient contingency planning can paralyze an entire fleet during foreseeable disruptions. Each of these factors alone, and especially in combination, can lead to accidents.

Self-driving car companies may be repeating many of the same early mistakes made in military UAV programs more than 30 years ago. Reliance on remote human intervention with long latencies, combined with opaque staffing models, unclear training standards, and limited evidence of robust contingency planning, suggests that remote operations have been treated as an operational afterthought rather than a safety-critical subsystem. If self-driving vehicles are to scale safely in the real world, remote operations must not be trivialized and should be approached with the same rigor applied to flight-critical systems in aviation. The military UAV community learned these lessons at great cost. The self-driving industry has the opportunity— and the responsibility —to learn them faster, before similar failures cause public harm.

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